PROJECT SYNOPSIS

ThreatScope

A Reproducible SOC Detection Lab with Ground-Truth Rule Evaluation

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Course	Design Project – III · Semester V · 2025–26
Domain	Security operations · detection engineering
Repository	github.com/harshacreate/ThreatScope

Introduction

ThreatScope is a compact, reproducible SOC lab: multi-source log and PCAP analysis, ATT&CK-mapped alerts with playbooks, isolated real tools, and a ground-truth benchmark for every rule.

Problem Statement

Production SIEMs are heavy and opaque for learners, detection rules are usually tuned without ground truth, and active scanning tools must never reach third-party systems.

Objectives

- Parse four log types; five sliding-window rules mapped to ATT&CK
- TShark and Suricata PCAP analysis
- Nmap and ZAP restricted to an internal demo target
- Ground-truth benchmark with threshold sweeps
- Analyst console; hardened and tested platform

Methodology

Synthetic generator with injected incidents; typed parsers; deque-based windows; Compose with an internal lab network; benchmark over 25 seeds with precision/recall; Flask, Docker, pytest.

Tools

Python, Flask, scapy, TShark, Suricata, Nmap, OWASP ZAP, Docker Compose, pytest, GitHub Actions.

Expected Outcomes

- Working SOC console with real tools
- Measured rule quality and operating bands
- Report, presentation, demonstration

References

- MITRE ATT&CK (Strom et al., 2018)
- NIST SP 800-61r2; NIST SP 800-92
- Sommer & Paxson, IEEE S&P 2010
- Suricata, Nmap and ZAP documentation